

Machine Learning in Action Notes

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1 An Overview of Machine Learning

Machine Learning (ML) is a subset of AI focused on teaching computers to learn from data, identify patterns, and make decisions without explicit programming. AI includes fields like computer vision, natural language processing, generative AI, ML, and deep learning. Deep learning uses multi-layered neural networks to automatically extract features from complex data.

Types of ML:

- **Supervised Learning:** Trains on labeled data to predict outcomes (e.g., classification, regression).
- **Unsupervised Learning:** Finds patterns in unlabeled data (e.g., clustering, association).
- **Semi-supervised Learning:** Uses a small amount of labeled data and iteratively labels more data.
- **Reinforcement Learning:** Learns by interacting with an environment and receiving feedback.

Common ML Techniques:

- **Classification:** Predicts discrete categories (e.g., benign vs. malignant cells).
- **Regression:** Predicts continuous values (e.g., house prices).
- **Clustering:** Groups similar data points (e.g., customer segmentation).
- **Association:** Finds items/events that co-occur (e.g., market basket analysis).
- **Anomaly Detection:** Identifies unusual cases (e.g., fraud detection).
- **Sequence Mining:** Predicts next events (e.g., clickstream analysis).
- **Dimensionality Reduction:** Reduces data size/features.
- **Recommendation Systems:** Suggests items based on user preferences.

Applications: Disease prediction, consumer behavior analysis, image recognition, recommendation systems (Amazon, Netflix), loan approval, customer churn prediction, computer vision, chatbots, and more.

Key Points:

- ML models require data preparation and algorithm selection.
- Human oversight remains important for interpreting and acting on ML outputs.
- ML impacts many industries and daily life.